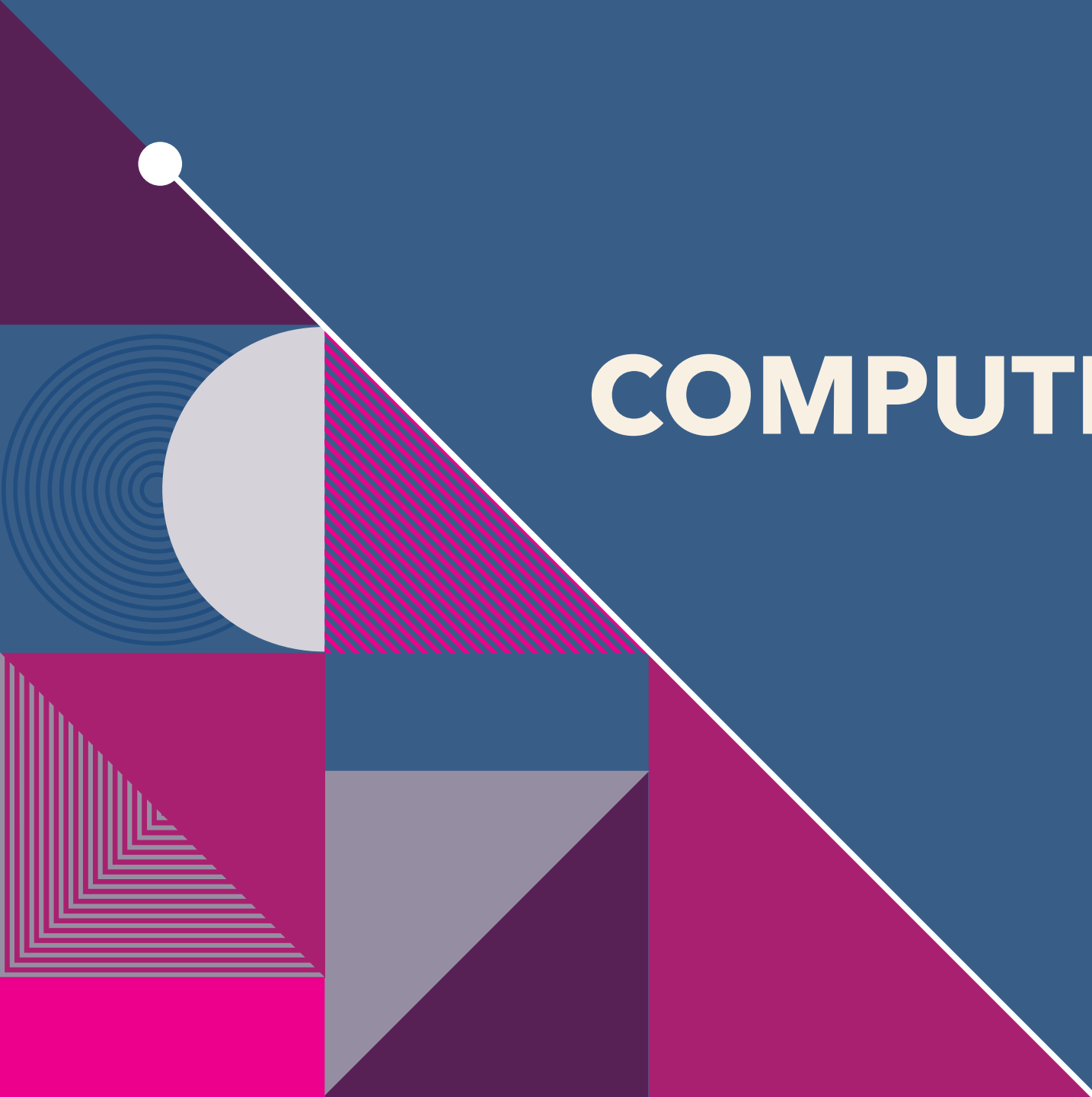
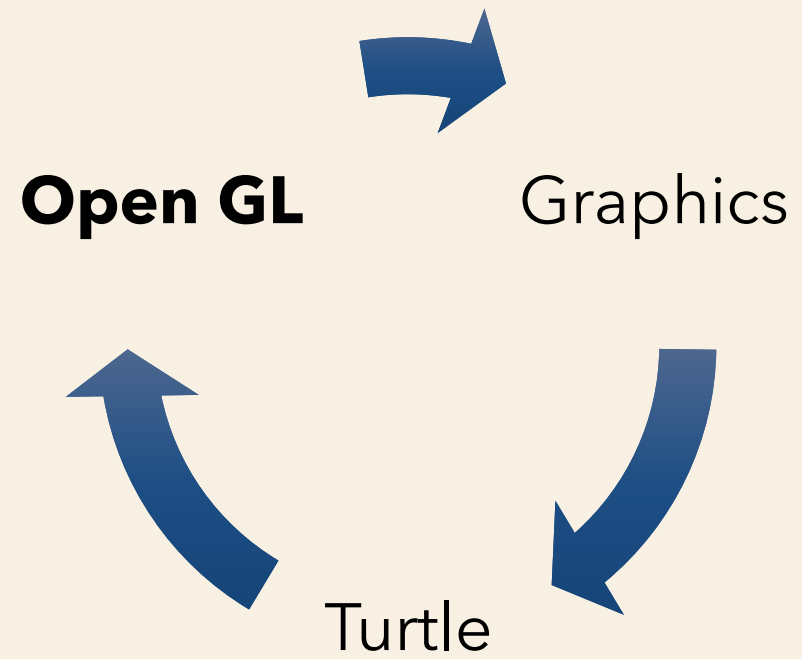




COMPUTER GRAPHICS SEC8

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LIBRARIES



MOUSE & KEYBOARD FUNCTION EXERCISES

Change Background Color

```
from OpenGL.GL import *
from OpenGL.GLUT import *
from OpenGL.GLU import *
```

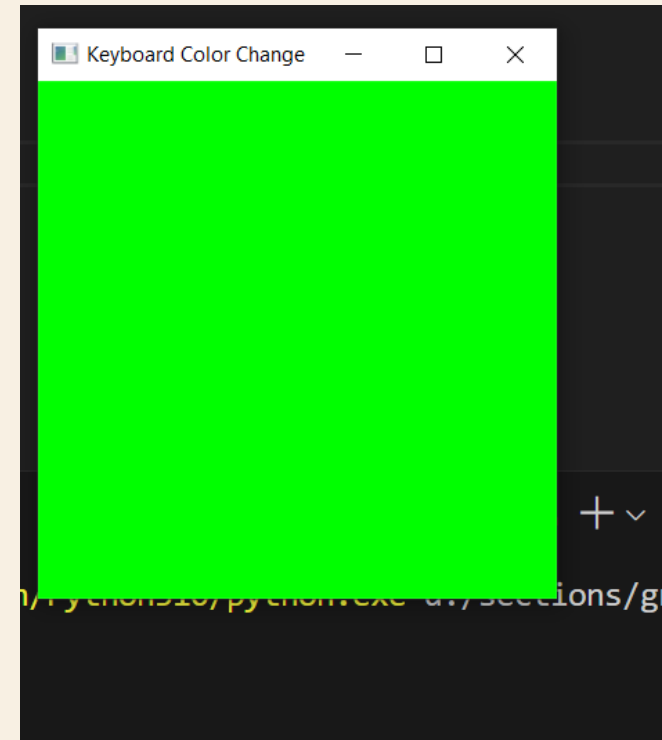
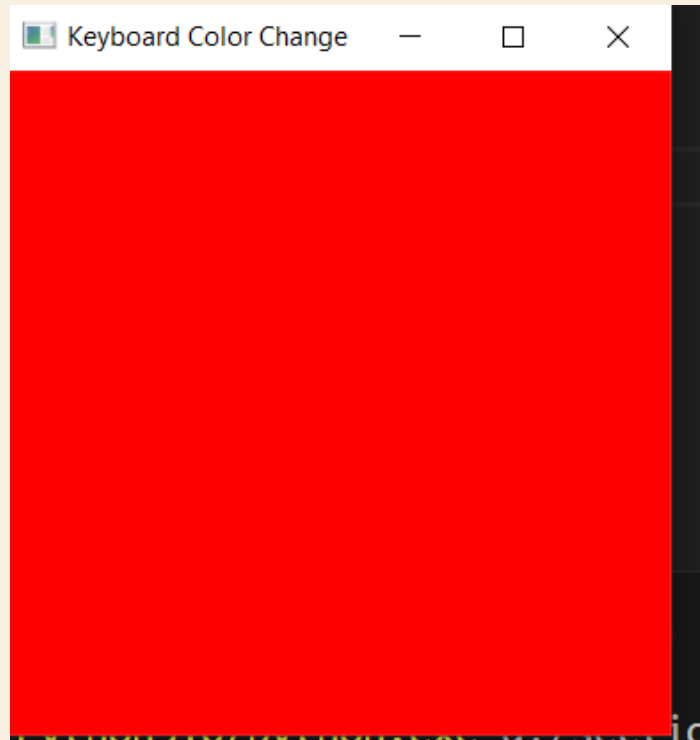
```
bg_color = [0.0, 0.0, 0.0]
```

```
def keyboard(key, x, y):
    global bg_color
    if key == b'r':
        bg_color = [1.0, 0.0, 0.0]
    elif key == b'g':
        bg_color = [0.0, 1.0, 0.0]
    elif key == b'b':
        bg_color = [0.0, 0.0, 1.0]
    glutPostRedisplay()
```

```
def display():
    glClearColor(*bg_color, 1.0)
    glClear(GL_COLOR_BUFFER_BIT)
    glFlush()

glutInit()
glutCreateWindow(b"Keyboard Color Change")
glutDisplayFunc(display)
glutKeyboardFunc(keyboard)
glutMainLoop()
```

MOUSE & KEYBOARD FUNCTION EXERCISES



MOUSE CLICK - DRAW POINTS

```
points = []

def mouse(button, state, x, y):
    if state == GLUT_DOWN:
        if button == GLUT_LEFT_BUTTON:
            points.append((x, 480 - y)) # convert to OpenGL coords
        elif button == GLUT_RIGHT_BUTTON:
            points.clear()
        glutPostRedisplay()

def display():
    glClear(GL_COLOR_BUFFER_BIT)
    glColor3f(1.0, 0.0, 0.0)
    glPointSize(5)
    glBegin(GL_POINTS)
    for x, y in points:
        glVertex2f(x, y)
    glEnd()
    glFlush()
```

```
def reshape(w, h):
    glViewport(0, 0, w, h)
    glMatrixMode(GL_PROJECTION)
    glLoadIdentity()
    gluOrtho2D(0, w, 0, h)

glutInit()
glutInitWindowSize(640, 480)
glutCreateWindow(b"Mouse Draw")
glutDisplayFunc(display)
glutMouseFunc(mouse)
glutReshapeFunc(reshape)
glutMainLoop()
```



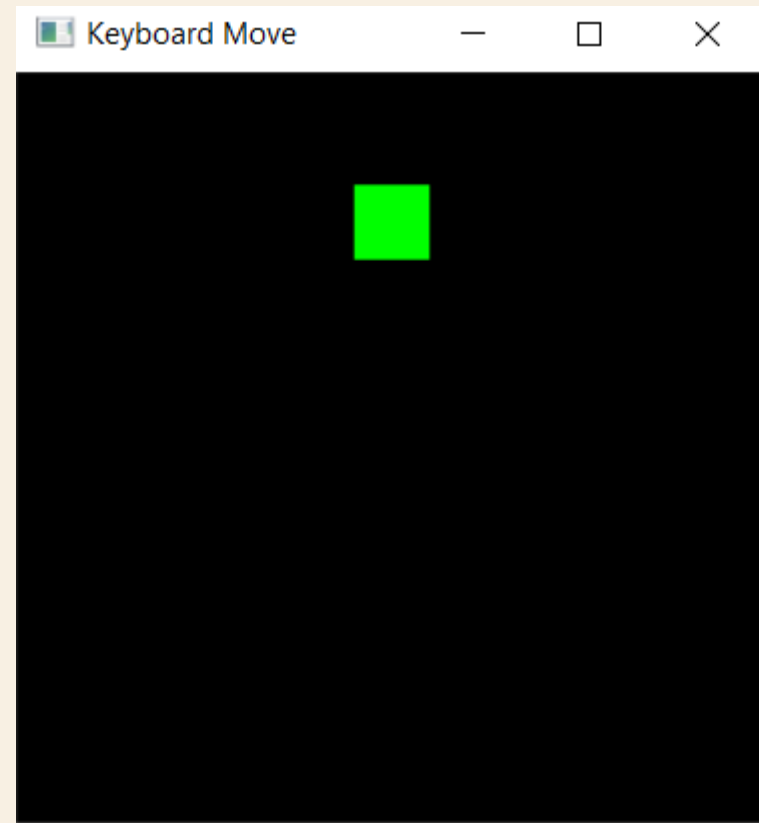
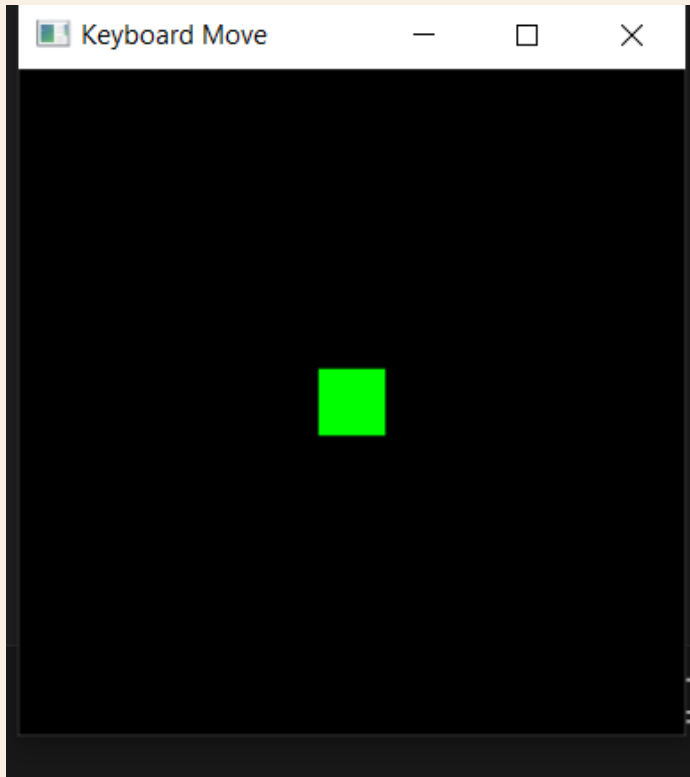
KEYBOARD MOVEMENT - MOVE A SQUARE

```
pos_x, pos_y = 0.0, 0.0

def keyboard(key, x, y):
    global pos_x, pos_y
    step = 0.1
    if key == b'w':
        pos_y += step
    elif key == b's':
        pos_y -= step
    elif key == b'a':
        pos_x -= step
    elif key == b'd':
        pos_x += step
    glutPostRedisplay()
```

```
def display():
    glClear(GL_COLOR_BUFFER_BIT)
    glLoadIdentity()
    glTranslatef(pos_x, pos_y, 0)
    glColor3f(0.0, 1.0, 0.0)
    glBegin(GL_QUADS)
    glVertex2f(-0.1, -0.1)
    glVertex2f( 0.1, -0.1)
    glVertex2f( 0.1,  0.1)
    glVertex2f(-0.1,  0.1)
    glEnd()
    glFlush()

glutInit()
glutCreateWindow(b"Keyboard Move")
glutDisplayFunc(display)
glutKeyboardFunc(keyboard)
glutMainLoop()
```



DRAW OBJECT WITH MOUSE MOTIONPYTHONCOPYEDIT

```
dragging = False
x, y = 300, 300

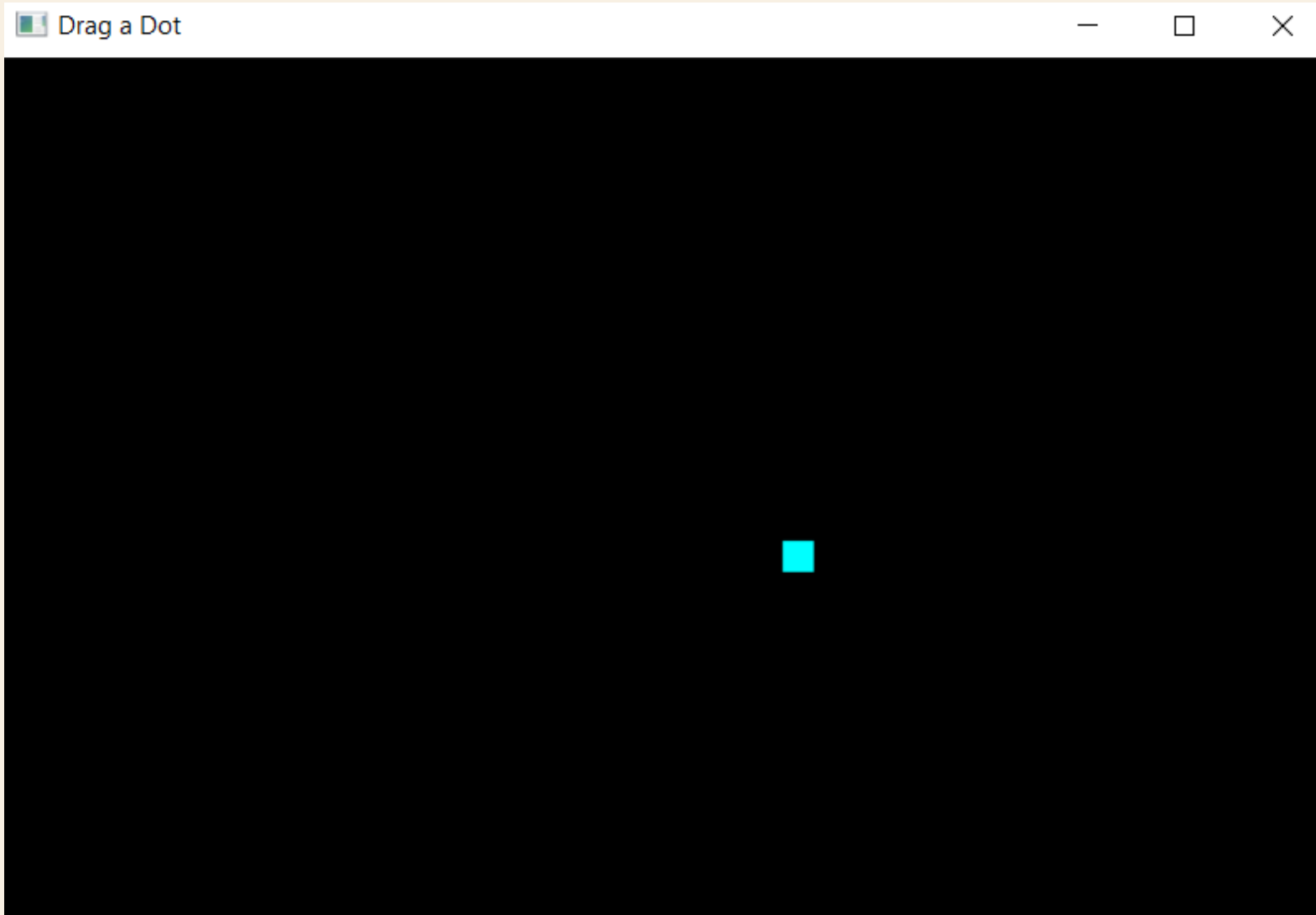
def mouse(button, state, mx, my):
    global dragging
    if button == GLUT_LEFT_BUTTON:
        dragging = (state == GLUT_DOWN)

def motion(mx, my):
    global x, y
    if dragging:
        x = mx
        y = 480 - my
    glutPostRedisplay()
```

```
def display():
    glClear(GL_COLOR_BUFFER_BIT)
    glColor3f(0.0, 1.0, 1.0)
    glPointSize(15)
    glBegin(GL_POINTS)
    glVertex2f(x, y)
    glEnd()
    glFlush()

def reshape(w, h):
    glViewport(0, 0, w, h)
    glMatrixMode(GL_PROJECTION)
    glLoadIdentity()
    gluOrtho2D(0, w, 0, h)

glutInit()
glutInitWindowSize(640, 480)
glutCreateWindow(b"Drag a Dot")
glutDisplayFunc(display)
glutMouseFunc(mouse)
glutMotionFunc(motion)
glutReshapeFunc(reshape)
glutMainLoop()
```



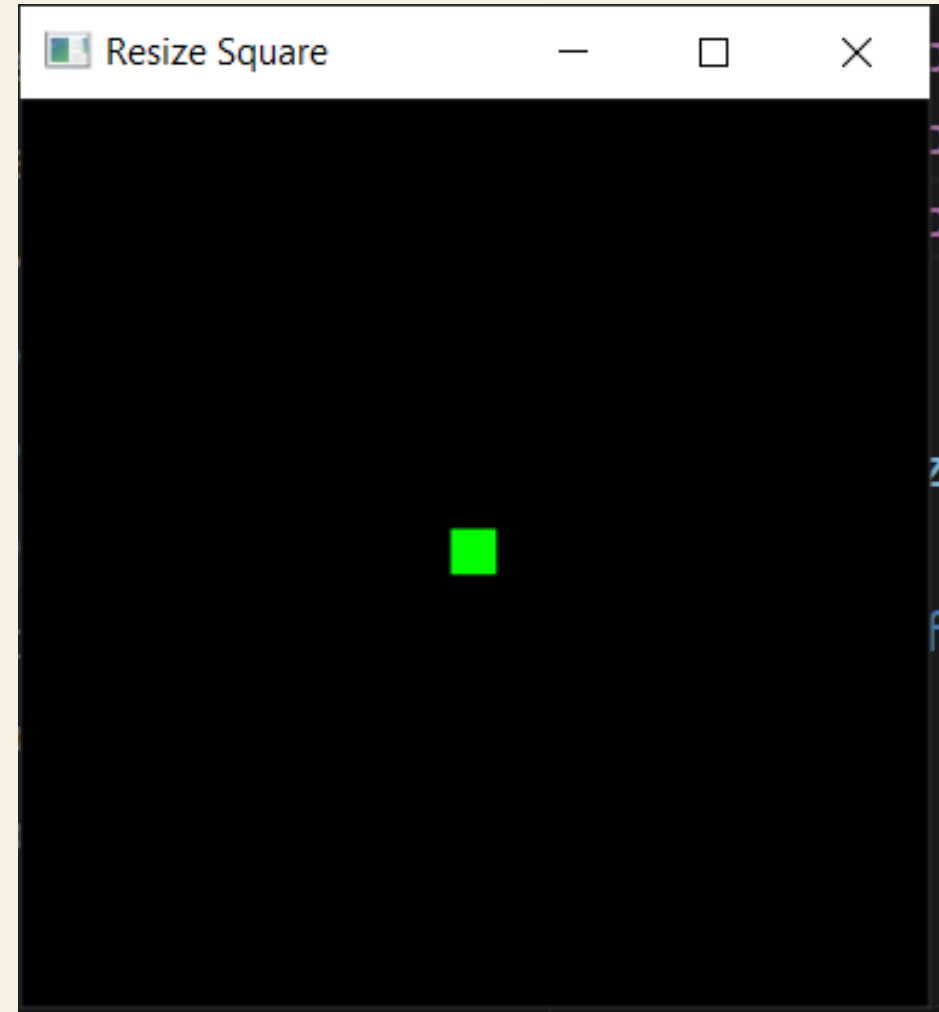
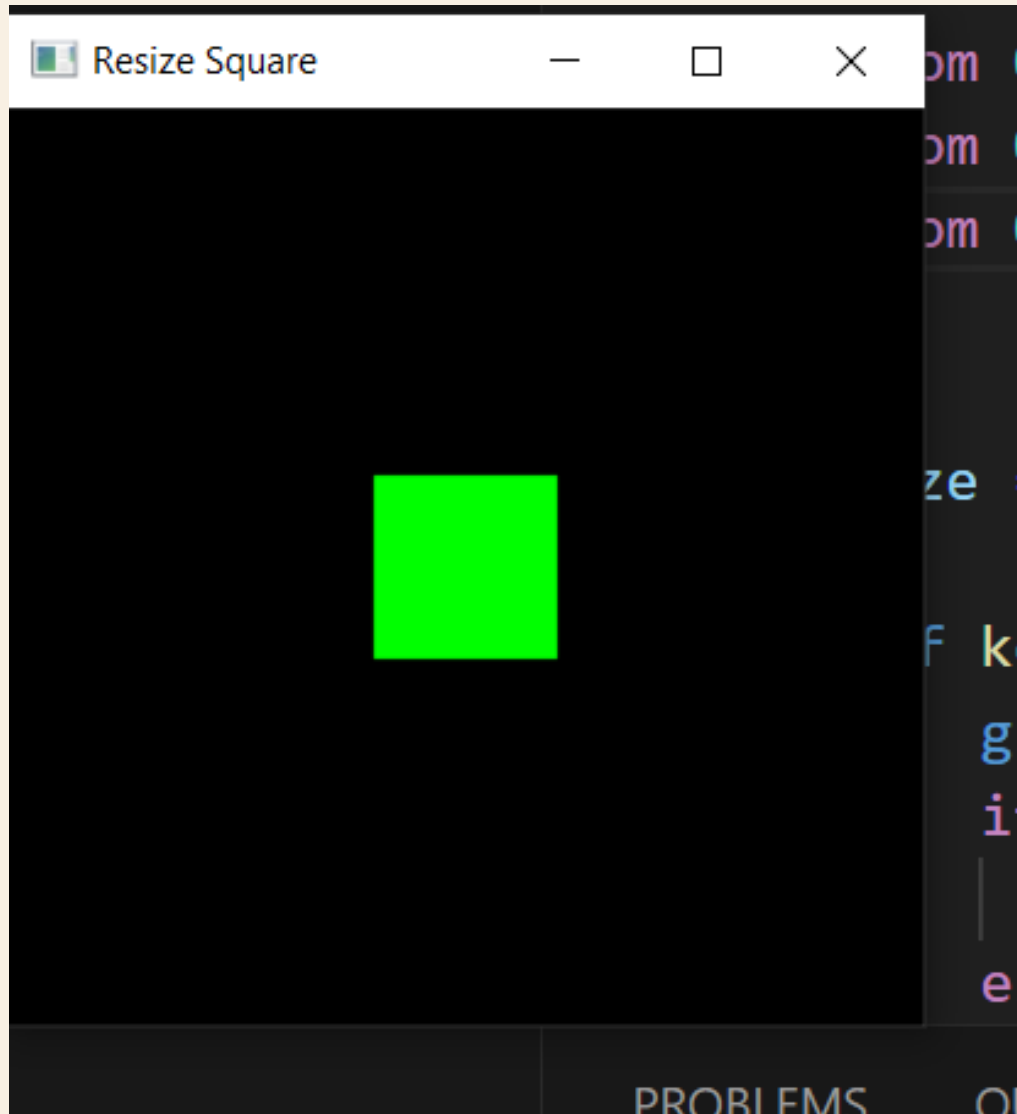
RESIZE OBJECT USING + / - KEYS

```
size = 0.2

def keyboard(key, x, y):
    global size
    if key == b'+':
        size += 0.05
    elif key == b'-':
        size = max(0.05, size - 0.05)
    glutPostRedisplay()
```

```
def display():
    glClear(GL_COLOR_BUFFER_BIT)
    glColor3f(0.0, 1.0, 0.0)
    glBegin(GL_QUADS)
    glVertex2f(-size, -size)
    glVertex2f( size, -size)
    glVertex2f( size,  size)
    glVertex2f(-size,  size)
    glEnd()
    glFlush()

glutInit()
glutCreateWindow(b"Resize Square")
glutDisplayFunc(display)
glutKeyboardFunc(keyboard)
glutMainLoop()
```



TASK

Draw Line Between Two Clicks